

1 Terms of Reference

1.1 Project Title

An investigation into unsupervised machine Learning for predicting customer segment: Autoencoder method.

1.2 Project Overview

Aim: This project aims to explore the application of unsupervised machine learning, specifically autoencoder, to predict customer segments based on their demographic and behavioural characteristics. The study will investigate the effectiveness of autoencoder over traditional methods in identifying distinct customer groups and providing insights for targeted marketing strategies.

Objectives:

- Conduct a comprehensive literature review and analyse at least 15 peer-reviewed articles on unsupervised machine learning techniques for customer segmentation, focusing on autoencoder methods.
- To develop an autoencoder model: Design, train, and test an autoencoder model using demographic and behavioural data from a dataset of at least 1000 customers, achieving a minimum accuracy of 85% in identifying distinct customer segments
- To evaluate the performance of the autoencoder model: Assess the performance of the autoencoder model using metrics such as silhouette score, calinski-harabasz index, and davies-bouldin index, and compare the results with traditional clustering algorithms
- To compare the performance of the autoencoder model with at least 2 traditional clustering algorithms (e.g., k-means, hierarchical clustering) using the same dataset, and quantify the improvement in identifying distinct customer segments
- Provide recommendations for businesses: Based on the findings, develop a report outlining the benefits and limitations of using autoencoder for customer segmentation, and provide actionable recommendations for businesses on how to apply autoencoder for targeted marketing strategies

Research Question:

How can autoencoder-based unsupervised machine learning be effectively utilized to identify and predict customer segments, and what advantages does it offer over traditional clustering methods?

Practical outcomes:

- A preprocessed and cleaned dataset suitable for training the autoencoder model on customer data.

- A developed and trained autoencoder model that can accurately identify distinct customer segments based on demographic and behavioral characteristics.
- A Python code and dataset used for the study, made available for other researchers and practitioners to replicate and build upon the results.
- A comparative analysis of the autoencoder's segmentation results against traditional clustering algorithms.

1.3 Underpinning research with Literature Review

Citation	Brief summary of paper	Relevance to your research question	Relevance to practical outcome of project
Alkhayrat, M., Aljnidi, M. and Aljoumaa, K. (2020). A comparative dimensionality reduction study in telecom customer segmentation using deep learning and PCA. Journal of Big Data, 7(1). doi: https://doi.org/10.1186/s40537-020-0286-0 .	The paper proposes using an autoencoder neural network for dimensionality reduction on a large telecom customer dataset, before applying k-means clustering. It compares the autoencoder approach to Principal Component Analysis (PCA) for dimensionality reduction. The autoencoder exhibited better clustering performance than PCA, suggesting its effectiveness in extracting relevant features for customer segmentation	The paper is relevant as it explores the application of autoencoder-based unsupervised learning for customer segmentation, highlighting its effectiveness in dealing with high-dimensional data and comparing it against traditional clustering methods such as PCA and k-means.	The study provides a comprehensive approach to preprocessing and cleaning datasets, developing a robust autoencoder model, and optimizing its architecture and hyperparameters. It also includes a comparative analysis of segmentation results from autoencoders and traditional methods, directly addressing the practical outcomes of data preparation, model development, and performance evaluation.

Höchst, J., Lars Baumgärtner, Hollick, M. and Freisleben, B. (2017). Unsupervised Traffic Flow Classification Using a Neural Autoencoder. doi:https://doi.org/10.1109/lcn.2017.57.	The paper presents an unsupervised traffic flow classification approach using a neural autoencoder. It constructs feature vectors from statistical properties of network flows over exponentially growing time intervals. The autoencoder clusters the flows into different classes like downloads, uploads, calls, etc., without relying on predefined traffic classes. Evaluation on real data shows the approach can detect 7 different traffic classes with 80% precision and 75% recall.	The paper demonstrates how an autoencoder can be effectively used for unsupervised clustering and identification of different segments/classes in network traffic data, without relying on labelled data. This relates to using autoencoders for customer segmentation in an unsupervised manner.	The paper provides insights into preprocessing network traffic data into suitable feature vectors for the autoencoder model. It also highlights the importance of exploring different autoencoder architectures, hyperparameters, and scaling methods to optimize performance. The comparative analysis against traditional clustering methods like K-means can guide the evaluation of the autoencoder's segmentation results.
Mancisidor, R.A., Kampffmeyer, M., Aas, K. and Jenssen, R. (2021). Learning latent representations of bank customers with the Variational Autoencoder. Expert Systems with Applications, 164, p.114020. doi:https://doi.org/10.1016/j.eswa.2020.114020.	The paper explores the use of Variational Autoencoders (VAEs) for credit scoring and customer segmentation in the financial industry. It highlights the advantages of VAEs in capturing non-linear relationships and generating meaningful clusters in the latent space. The research demonstrates that segment-based credit scoring using VAEs outperforms traditional portfolio-based methods.	The paper is relevant as it showcases the effectiveness of autoencoder-based unsupervised learning in identifying and predicting customer segments, emphasizing the advantages over traditional clustering methods like improved performance and the ability to handle non-linear data relationships.	This research is directly applicable to the project outcomes as it provides a methodology for preprocessing customer data, developing and optimizing an autoencoder model, and conducting a comparative analysis with traditional clustering techniques. The findings support the practical implementation of VAEs for more accurate and insightful

			customer segmentation in a financial context.
Nguyen, S.P. (2021). Deep customer segmentation with applications to a Vietnamese supermarkets' data. <i>Soft Computing</i>, 25(12), pp.7785–7793. doi:https://doi.org/10.1007/s00500-021-05796-0.	This paper focuses customer segmentation in Vietnamese supermarkets using soft computing and data mining techniques. The authors proposed a new method for customer segmentation that utilizes machine learning algorithms and data analysis to group customers into different segments based on their purchasing behaviour and other factors. The paper also discusses the potential benefits of this approach for personalized marketing and improving customer satisfaction.	The paper is highly relevant to the research question as it specifically addresses the utilization of autoencoder-based unsupervised machine learning for customer segmentation. It provides insights into the effectiveness of autoencoders in identifying and predicting customer segments and offers a comparative analysis against traditional clustering methods, thus helping to understand the advantages offered by this approach.	The preprocessing and cleaning techniques outlined in the paper will guide in preparing the dataset for model training. The development and optimization of the autoencoder model provide a framework that can be adapted specific customer data.
Salehinejad, H. and Rahnamayan, S. (2016). Customer shopping pattern prediction: A recurrent neural network approach. 2016 IEEE Symposium Series on Computational Intelligence (SSCI). doi:https://doi.org/10.1109/ssci.2016.7849921.	The paper proposes a recurrent neural network (RNN) model with an autoencoder component to predict customer shopping behaviour based on recency, frequency, and monetary (RFM) values. The autoencoder extracts features from the customer ID and RFM values, which are fed into the RNN to predict future RFM values.	The paper discusses the use of autoencoder-based unsupervised learning for customer segmentation, which aligns with the research question on identifying and predicting customer segments. It highlights how feature extraction through autoencoders can improve prediction accuracy compared	Preprocessed and Cleaned Dataset: The paper provides insights into the preprocessing steps necessary for preparing transactional data for training the autoencoder model, including one-hot encoding for customer loyalty numbers. Development of a Robust Autoencoder

	Experiments show the RNN with rectified linear units performed best for RFM prediction compared to simple RNNs and LSTMs.	to traditional methods.	Model: It details the architecture and training process of an autoencoder integrated within an RNN model, demonstrating its application in customer segmentation. Optimized Model Architecture and Hyperparameters: The paper outlines the specific architecture and hyperparameters used, such as the number of hidden units, epochs, and autoencoder features, which can guide the optimization of the model.
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Smutek, T., Sikora, J., Bogacki, S., Rutkowski, M. and Woźniak, D., 2024, Use of Autoencoder and One-Hot Encoding for Customer Segmentation.	The research paper applies Autoencoder and One-Hot Encoding techniques to segment retail customers based on their profiles. It evaluates the effectiveness of these methods using various clustering algorithms like K-means, DBSCAN, Louvain Community Detection, Greedy Modularity, and Label Propagation. The study finds that the Louvain method with AutoEmbedder produces more cohesive and meaningful customer segments compared to One-Hot Encoding, as evidenced by higher modularity scores.	The paper directly addresses the utilization of autoencoder-based unsupervised machine learning for customer segmentation. It demonstrates the advantages of autoencoders over traditional One-Hot Encoding by forming more modular and cohesive customer clusters. The comparative analysis with established clustering algorithms highlights the potential benefits of using autoencoders for this task.	The research provides a practical approach to preprocess customer data and train an autoencoder model for segmentation. It offers insights into optimizing the autoencoder architecture and evaluating its performance against benchmarks. The clustering quality metrics used can guide the selection of appropriate algorithms and parameters for effective customer segmentation in practical applications.
Soni, J., Prabakar, N., Upadhyay, H. (2023). Deep Learning-Based Efficient Customer Segmentation for Online Retail Businesses. In: Yang, XS. (eds) Benchmarks and Hybrid Algorithms in Optimization and Applications. Springer Tracts in Nature-Inspired Computing. Springer, Singapore. https://doi.org/10.1007/978-981-99-3970-1_9	The paper discusses the importance of customer segmentation in today's competitive market and proposes a learning-based framework using unsupervised machine learning algorithms, specifically K-Means and AutoEncoders, to segment customers. The framework is implemented on an open-source dataset from Kaggle, and the results are compared with traditional clustering methods. The paper	The paper directly addresses the research question by exploring the effectiveness of AutoEncoder-based unsupervised machine learning in identifying and predicting customer segments. It provides a comprehensive analysis of the advantages of Autoencoders over traditional clustering methods, thereby answering the research question.	The paper aligns with the practical outcomes by providing a preprocessed and cleaned customer dataset, developing a robust autoencoder model for segmentation, optimizing the model architecture and hyperparameters, and conducting a comparative analysis of the autoencoder's segmentation results against traditional clustering

	aims to provide an effective way to identify and predict customer segments using Autoencoder-based unsupervised machine learning.		algorithms. These outcomes can directly contribute to the practical implementation of customer segmentation using deep learning techniques.
Tavakoli, N., Siامي-Namini, S., Khanghah, M.A., Soltani, F.M. and Namin, A.S. (2020). Clustering Time Series Data through Autoencoder-based Deep Learning Models. [online] arXiv.org. doi: https://doi.org/10.48550/arXiv.2004.07296 .	The paper introduces an autoencoder-based synergic methodology for time series clustering, specifically focusing on financial data. It details the architecture of an encoder-decoder neural network designed to learn features from time series data, emphasizing the ability to capture salient features and reconstruct original inputs for clustering.	The paper is directly relevant as it explores how autoencoder-based unsupervised learning can effectively cluster time series data, demonstrating the advantages of capturing hidden features over traditional clustering methods.	This paper provides detailed insights into the development and optimization of an autoencoder model, including architecture and hyperparameters, for clustering tasks. The comparative analysis with traditional clustering methods aligns with the project's goals of evaluating the efficacy of autoencoder-based segmentation.
Wang, C. (2022). Efficient customer segmentation in digital marketing using deep learning with swarm intelligence approach. Information Processing & Management, 59(6), p.103085. doi: https://doi.org/10.1016/j.ipm.2022.103085 .	The paper proposes an unsupervised deep learning model that combines a self-organizing map (SOM) with a modified social spider optimization (MSSO) algorithm for customer segmentation in digital marketing. Features are selected using MSSO, and the SOM clusters customers based on their purchasing patterns. The clustered data is then classified using a	The paper is highly relevant as it explores the use of an autoencoder-based unsupervised deep learning model (SOM) for customer segmentation. It demonstrates how the autoencoder can effectively identify and group customers based on their purchasing behaviour, which is a key aspect of the research question on utilizing autoencoders for	The paper provides a practical approach to preprocess and clean the customer dataset using feature selection techniques like MSSO. It also outlines the development of the SOM autoencoder model for segmentation and optimizes its architecture through quantization and topological error analysis.

	deep neural network (DNN) to segment customers into different groups for targeted marketing strategies.	customer segmentation.	Additionally, it compares the autoencoder's performance with traditional clustering methods, fulfilling the practical outcomes outlined in the project.
Wu, H. and Wang, Q. (2022). GEAE: Gated Enhanced Autoencoder based Feature Extraction and Clustering for Customer Segmentation. 2022 International Joint Conference on Neural Networks (IJCNN). doi: https://doi.org/10.1109/ijcnn55064.2022.9892475 .	The paper introduces the Gated Enhanced Autoencoder (GEAE) model, which enhances clustering performance and reconstruction error compared to traditional methods. It highlights the importance of capturing nonlinear features in data and the impact of position selection of the gating block in improving model performance. The study focuses on customer segmentation using autoencoders for dimensionality reduction and emphasizes the significance of efficient feature extraction for clustering accuracy.	The paper is relevant to the research question as it by proposed Gated Enhanced Autoencoder (GEAE) model for customer segmentation. It demonstrates how autoencoder-based unsupervised learning can efficiently reduce data dimensions and improve clustering accuracy by capturing complex features. It show how autoencoders could be utilised for the study.	Gives fundamental understanding of how to performed segmentation for customer using autoencoder alongside factors to consider in the data preprocessing and background understanding.

2 Project Schedule

2.1 Table 1: Effort

Task Id	Task Name	Start	Deadline	Hour	Deliverable
1.0	Project initiation and Literature Review				
1.1	Project aim and objective declaration	09/06/2024	09/06/2024	6	Draft of chapter one
1.1	Identify relevant publications	10/06/2024	10/06/2024	4	List of relevant publications
1.2	Analyze and synthesize findings	11/06/2024	11/06/2024	7	Summary of key findings
1.3	Write literature review report	12/06/2024	13/06/2024	15	Literature review report
1.4	Completion of chapter 1	14/06/2024	15/06/2024	15	Complete report of chapter 1
2.0	Dataset Preparation				
2.1	Sourcing for possible dataset	16/06/2024	16/06/2024	6	List of identified possible datasets to use
2.2	Collection of open-source customer dataset	17/06/2024	18/06/2024	10	Selected raw dataset
2.3	Preprocessing of collected dataset	19/06/2024	20/06/2024	12	Preprocessed and annotated dataset
2.4	Documentation of chapter two	21/06/2024	22/06/2024	15	Complete report of chapter 2
3.0	Model Development				
3.1	Identification and selection of research methodology	23/06/2024	24/06/2024	2	Summary of the research methodology
3.2	Exploration autoencoders for segmentation	25/06/2024	25/06/2024	3	List of combination of different uses of autoencoder for segmentation
3.3	Python code implementation of the selected algorithm	26/06/2024	28/06/2024	15	Code artifact of the implementation
3.4	Training of the autoencoder model	29/06/2024	03/07/2024	23	Trained deep learning models
3.4	Testing of trained model	04/07/2024	04/07/2024	2	Performance report of model
3.5	Refine and optimize models	05/07/2024	05/07/2024	5	Refined and optimized models
3.4	Documentation of chapter 3	06/07/2024	09/07/2024	15	Complete report of chapter 3
4.0	Evaluation and Benchmarking				
4.1	Gathering of research result	10/07/2024	10/07/2024	4	First stage summary of the research result

4.1	Define evaluation metrics	11/07/2024	12/07/2024	1.5	Evaluation metrics and benchmarks
4.2	Evaluate model performance	13/07/2024	14/07/2024	4	Performance evaluation results
4.3	Benchmark against other methods	15/07/2024	16/07/2024	3	Benchmarking report
5.0	Evaluation and Reporting				
5.1	Completion of chapter 4	17/07/2024	19/07/2024	15	Complete report of chapter 4
5.2	Identification if objectives have been met	20/07/2024	22/07/2024	9	Refined and achievement table
6.0	Documentation				
6.1	Conclusion documentation	23/07/2024	25/07/2024	15	Complete report on chapter 5
6.2	Finalize project documentation	26/07/2024	04/08/2024	25	Completed project documentation
6.3	Preparation for Viva	05/08/2024	06/08/2024	3	Complete PowerPoint presentation
6.4	Submissions of report and artifacts	10/08/2024	10/08/2024	1	Submitted report and artifacts

2.2 Table 2: Deliverables

Del. No.	Name	Deadline
D1	Complete report of chapter 1	15/06/2024
D2	Preprocessed and annotated dataset	20/06/2024
D3	Complete report of chapter 2	22/06/2024
D4	Summary of the research methodology and exploration of autoencoders for segmentation	25/06/2024
D5	Code artifact and trained deep learning models	03/07/2024
D6	Performance report and refined/optimized models	05/07/2024
D7	Complete report of chapter 3	09/07/2024
D8	Performance evaluation results and benchmarking report	16/07/2024
D9	Complete report of chapter 4	19/07/2024
D10	Finalized project documentation and complete PowerPoint presentation	06/08/2024
D11	Submitted report and artifacts	10/08/2024

2.3 Table 3: Milestones

Milestone	Name	Deadline	Evidence
M1	Project Initiation	09/06/2024	Project proposal, project plan
M2	Literature Review Completion	15/06/2024	Literature review report, chapter 1
M3	Dataset Preparation Completion	29/06/2024	Dataset preparation report, chapter 2
M4	Model Development Completion	16/07/2024	Model development report, chapter 3
M5	Evaluation and Benchmarking Completion	26/07/2024	Evaluation and benchmarking report, chapter 4

M6	Final Reporting Completion	01/08/2024	Final report, chapter 5
M7	Project Documentation Completion	05/08/2024	Completed project documentation
M8	Final Submission	10/08/2024	Submitted report and artifact

2.4 Table 4: Outline Schedule / Gantt chart

	June					July					August	
	1	2	3	4	5	6	7	8	9	10	11	12
Project Initiation	M1 D1											
Literature Review			M2									
Dataset Preparation				M3 D2								
Model Development						D4	M4 D5					
Evaluation and Benchmarking									M5			
Final Reporting										D9 M6		
Project Documentation											M7	
Final Submission												M8 D11

3 Evaluation Plan

Objective	Evaluation Approach	Evidence
Conduct a comprehensive literature review on unsupervised machine learning techniques for customer segmentation.	Systematic review of relevant publications, categorizing and synthesizing findings to identify gaps and opportunities for research contribution.	D1 and D2: List of relevant publications; Comprehensive literature review report. The list of relevant publications indicates thorough research into existing approaches, challenges, and gaps in the field. Additionally, the comprehensive literature review report showcases the synthesis of findings.
To develop an autoencoder model for customer segmentation using demographic and behavioral data.	Design and implement an autoencoder model, preprocess demographic and behavioral data, train and optimize the model for effective segmentation.	D3: Preprocessed and annotated dataset; D4: Code artifact and trained deep learning models. The preprocessed dataset ensures data quality, while the code artifact and trained models demonstrate the practical application and development of the autoencoder.
To evaluate the performance of the autoencoder model in	Use performance metrics such as accuracy, precision, recall, F1 score, and visualization techniques to	D5: Performance report and refined models. The performance report provides quantitative and qualitative

identifying distinct customer segments.	assess the effectiveness of the model.	assessments of the model's accuracy and effectiveness in segmenting customers.
To compare the results of the autoencoder model with traditional clustering algorithms.	Implement traditional clustering algorithms (e.g., K-means, hierarchical clustering) and compare their results with those of the autoencoder model.	D6: Evaluation results and benchmarking report. This report includes comparative analyses highlighting the strengths and weaknesses of the autoencoder model relative to traditional methods.
To provide recommendations for businesses on the application of autoencoder for customer segmentation.	Synthesize findings from the literature review, model development, and performance evaluation to formulate actionable business recommendations.	D7: Finalized project documentation and presentation. The documentation includes detailed recommendations, while the presentation highlights key insights for business stakeholders.

4 Social, Ethical, Legal and Professional issues

4.1 Social, Ethical, Legal and Professional Issues Table

Social issues	Bias and discrimination in customer segmentation, privacy concerns with collecting and using customer data which will be mitigated by using open-source data. Additionally, they may be unintended consequences or misuse of customer segmentation models.
Ethical issues	Potential for perpetuating historical biases or unfair treatment, balancing commercial interests with ethical considerations and ensuring algorithmic oversight and alignment with societal values.
Professional issues	Ensuring proper training and expertise in developing and deploying AI systems. Maintaining professional standards and ethical practices in AI development. Responsible AI practices, including bias testing, model auditing, and stakeholder engagement.
Legal Issues	Compliance with data privacy regulations (e.g., GDPR, CCPA), Furthermore, fairness and non-discrimination laws in customer segmentation and targeting. Ensuring Intellectual property rights and ownership of AI models, customer data and research papers utilized.